SCIENCE OLYMPIAD FOUNDATION

Class 7

IMO

Set A

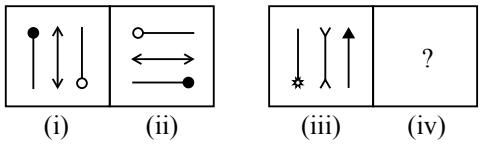
Year 2019

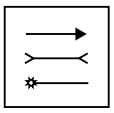
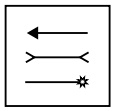
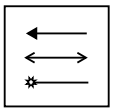
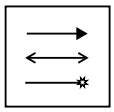
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1. If the digits '5' and '8', signs '+' and '×' are interchanged on the left hand side in each option, then which of the following becomes correct?

   A. $5 + 8 \times 2 = 22$
   B. $8 + 5 \times 2 = 18$
   C. $5 \times 8 + 2 = 42$
   D. $8 \times 2 + 5 = 21$

2. There is a certain relationship between figures (i) and (ii). Establish a similar relationship between figures (iii) and (iv) by selecting a suitable figure from the options which will replace the (?) in figure (iv).

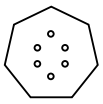
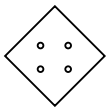
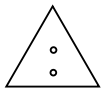


A. 
B. 
C. 
D. 

3. Kirti is the daughter of Smriti. Amit is the son of Smriti's brother Sahil. Smriti is the mother of Nisha. How is Kirti related to Nisha?

   A. Sister
   B. Mother
   C. Aunt
   D. None of these

4. Select the odd one out.

A. 
B. 
C. 
D. 

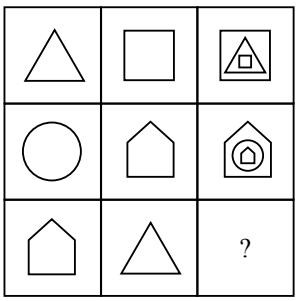
5. How many such pairs of letters are there in the word CENTURY each of which has as many letters between them in the word as there are in the English alphabet?

   A. One
   B. Two
   C. Three
   D. Four

6. Five boys took part in a race. Manish finished before Yash but after Gaurav. Avinash finished before Yogesh but after Yash. Who finished the race at second position?

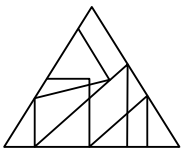
   A. Manish
   B. Gaurav
   C. Yash
   D. Yogesh

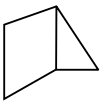
7. Which of the following options will complete the given figure matrix?



A. 
B. 
C. 
D. 

8. Select a figure from the options which is exactly embedded in the given figure as one of its parts.



A. 
B. 
C. 
D. 

9. If the middle digit of the greatest number is subtracted from the middle digit of the smallest number after adding one to each of the given numbers, then which of the following will be obtained?

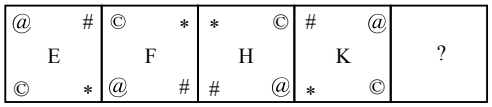
   718 612 525 312 297

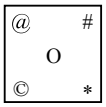
   A. 4
   B. 8
   C. 2
   D. 0

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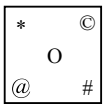
10. Select a figure from the options which will continue the same series as established by the Problem Figures.

Problem Figures



A. 

B. 

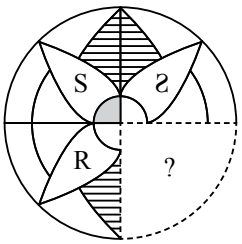
C. 

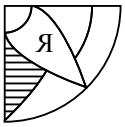
D. 

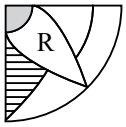
11. In a certain code language, 'REPAIR' is written as 'BJSSFQ'. How will 'SYSTEM' be written in the same language?

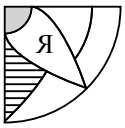
A. TZTUFN B. SDLRXR
C. RXRSDL D. UFNTZT

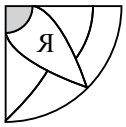
12. Select a figure from the options which will complete the pattern in the given figure.



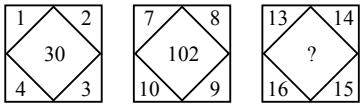
A. 

B. 

C. 

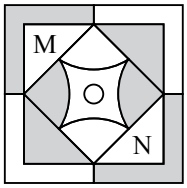
D. 

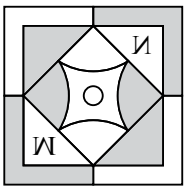
13. Find the missing number, if same rule is followed in all the three figures.

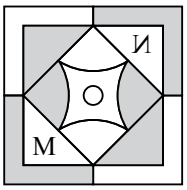


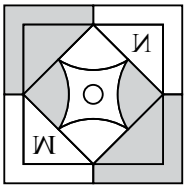
A. 58 B. 174
C. 116 D. 232

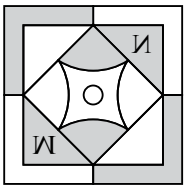
14. Select the correct water image of the given figure.



A. 

B. 

C. 

D. 

15. Ashina walked 30 m towards North. Then she turned right and walked 30 m. Again she turned right and walked 55 m. Then she turned left and walked 20 m. Finally she turned left and walked 25 m. In which direction is she now from her starting point?

A. West B. North
C. East D. South

MATHEMATICAL REASONING

16. If $\left(\frac{5}{2}\right)^5 \times \left(\frac{5}{2}\right)^{x+3} = \left(\frac{2}{5}\right)^{-10}$, then the value of x is

A. –1 B. 0
C. 10 D. 2

17. The given Fig. (X) is turned to a different position. Which of these CANNOT be the figure after it is turned ?

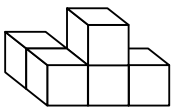
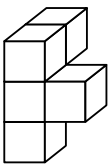
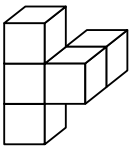
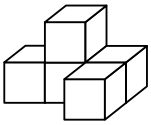
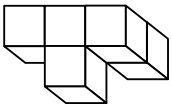


Fig. (X)

A. 

B. 

C. 

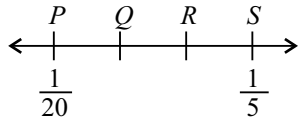
D. 

18. Find the value of 812 + 8.012 + 0.812 + 8.0012.

A. 825.3904 B. 728.7252
C. 828.9332 D. 828.8252

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19. The given number line shows four fractions *P, Q, R* and *S*. If *PQ* = *QR* = *RS*, then find the value of *Q* and *R* respectively.



A. $\frac{1}{4}, \frac{1}{5}$

B. $\frac{1}{10}, \frac{3}{20}$

C. $\frac{1}{10}, \frac{1}{5}$

D. $\frac{1}{5}, \frac{1}{10}$

---

20. ₹ 8000 have been divided into two parts such that if one part is lent at $7\frac{1}{2}\%$ per annum for 2 years and the other at 8% per annum for 3 years, then the total interest received is ₹ 1416. Find the two parts.

A. ₹ 5550, ₹ 2450

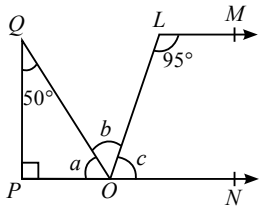
B. ₹ 5650, ₹ 2350

C. ₹ 5625, ₹ 2375

D. ₹ 5600, ₹ 2400

---

21. In the given figure (not drawn to scale), *LM* || *PN*. Find the value of *c* – *b*.



A. 55°

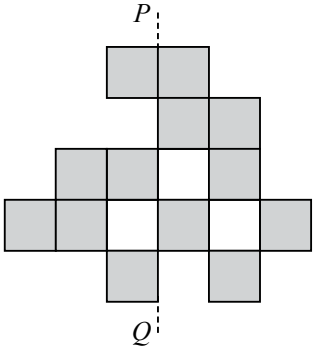
B. 35°

C. 40°

D. 30°

---

22. What is the least number of squares that must be added so that *PQ* becomes the line of symmetry?



A. 7

B. 8

C. 9

D. None of these

---

23. If $a = \frac{4lm}{l+m}$, then the value of $\frac{a+2l}{a-2l} + \frac{a+2m}{a-2m} =$

A. 2

B. 4

C. $\frac{2lma}{(l+m)}$

D. None of these

---

24. Which of the following is true for ΔABC?

A. $AB - BC > AC$

B. $AB + BC < AC$

C. $AB - BC < AC$

D. $AB + AC < BC$

---

25. Shikha has 18 red balls and 9 green balls. She chooses a ball randomly. Find the probability that the chosen ball is of red colour.

A. $\frac{4}{5}$

B. $\frac{1}{3}$

C. $\frac{3}{5}$

D. $\frac{2}{3}$

---

26. Find the value of *s* which makes the following equation true .

$s - 5 - \{5 - 4 + (7 \times 6 - 12 \div 3)\} = -40$

A. – 12

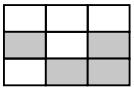
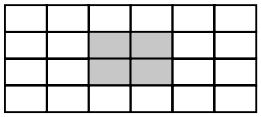
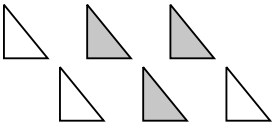
B. – 4

C. 4

D. 12

---

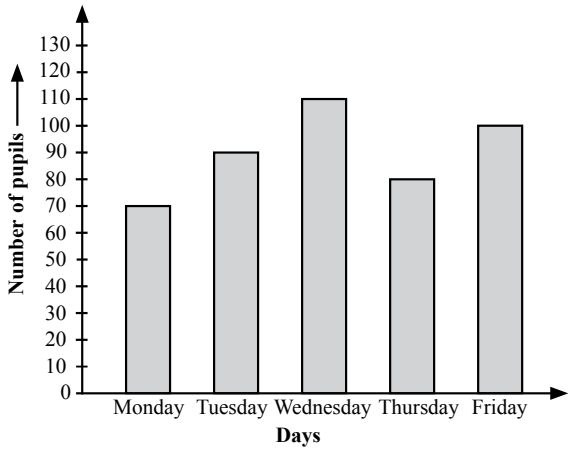
27. Select the correct match.

	Figure	**Fraction of unshaded part**
A.		$\frac{7}{9}$
B.		$\frac{1}{6}$
C.		$\frac{1}{2}$

D. None of these

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28. The given bar graph shows the number of pupils who participated in the race competition on five different days of a week.



What percentage of the total number of pupils participated in the race competition on Monday and Wednesday together?

A. 40%
B. 32%
C. 20%
D. 35%

---

29. $\frac{30\% \text{ of } 70\% \text{ of } 500}{35\% \text{ of } 45\% \text{ of } 400}$ is equal to _____ .

A. $\frac{5}{3}$
B. $\frac{3}{4}$
C. $\frac{1}{3}$
D. $\frac{1}{2}$

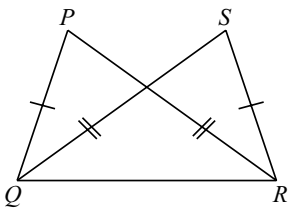
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30. Which of the following options is INCORRECT?

A. Two parallel lines never intersect each other.
B. Sum of all the angles around a point is 270°.
C. If two lines intersect, then the vertically opposite angles are equal.
D. Two adjacent supplementary angles form a linear pair.

---

31. If ΔPQR and ΔSQR are on the same base QR, $PQ = SR$ and $PR = SQ$, then which of the following gives a correct congruence relationship?



A. $\Delta PQR \cong \Delta SQR$
B. $\Delta PQR \cong \Delta RQS$
C. $\Delta PQR \cong \Delta SRQ$
D. $\Delta PQR \cong \Delta RSQ$

---

32. The heights (in cm) of 20 students of class VII are given below :

120, 115, 118, 140, 125, 126, 144, 140, 143, 136, 141, 138, 128, 140, 155, 144, 140, 152, 142, 156

Find

(a) The range of the given data.
(b) The mean height of the students.

	(a)	**(b)**
A.	26 cm	137 cm
B.	41 cm	137.15 cm
C.	152 cm	82 cm
D.	46 cm	140 cm

---

33. Select the INCORRECT step in simplification of the given equation.

$3(x + 2) = 4(2x - 1)$

Step-1 : $3x + 6 = 8x - 4$ (open brackets)

Step-2 : $8x - 3x = 6 + 4$ (combining like terms)

Step-3 : $5x = 15$

Step-4 : $x = 2$ (dividing by 5 on both sides)

A. Step-2
B. Step-4
C. Step-3
D. Step-1

---

34. In which of the following options, a unique triangle cannot be drawn?

A. $PQ = 6$ cm, $QR = 9$ cm and $RP = 4$ cm
B. $QR = 10$ cm, $\angle Q = 70^\circ$ and $\angle R = 60^\circ$
C. An equilateral triangle with the length of side 5 cm.
D. An isosceles triangle with the length of each equal side 4 cm.

---

35. A rectangular ground is 80 m long and 50 m broad. In the middle of the ground, there is a circular tank of radius 7 metres. Find the cost of turfing the remaining portion at the rate of ₹ 0.50 per square metre.

A. ₹ 2540
B. ₹ 1923
C. ₹ 3210
D. ₹ 4260

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36. Pooja walks $\frac{4}{5}$ km towards office from her home and then returns $\frac{2}{3}$ km on the same way towards her home to reach a landmark. How far will she be now from her home?

A. $\frac{1}{15}$ km

B. $\frac{1}{5}$ km

C. $\frac{2}{15}$ km

D. $\frac{2}{17}$ km

---

37. Varun scores 50 more runs than 5 times the runs scored by Karan. Form an equation which gives the runs scored by Karan, if Varun scored 120 runs.

A. $50x + 5 = 120$

B. $50 + 5x = 120$

C. $50x + 120 = 5$

D. $120 + 5x = 50$

---

38. In a test, 4 marks are given for every correct answer and –2 marks for every incorrect answer. Asha answered all the questions and scored 26 marks though she got nine correct answers. Varun also answered all the questions and scored 2 marks though he got five correct answers. How many questions did each of them answer incorrectly?

A. 10, 8

B. 9, 18

C. 5, 9

D. 4, 10

---

39. Four bells begin to ring together and ring respectively at an interval of 50 secs, 40 secs, 60 secs and 70 secs. After how much time will they again ring together?

A. 90 mins

B. 70 mins

C. 100 mins

D. 60 mins

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40. Shreya bought $5\frac{3}{4}$ litres of milk on Monday and $4\frac{2}{3}$ litres of milk on Tuesday. How many litres of milk did she buy together in two days?

A. $5\frac{3}{4}$ litres

B. $6\frac{1}{4}$ litres

C. $10\frac{5}{12}$ litres

D. $9\frac{3}{4}$ litres

---

41. A piggy bank contains an equal number of one rupee, 50 paise and 25 paise coins. If the total value is ₹ 56, then how many coins of each type are there?

A. 25

B. 32

C. 30

D. 22

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42. The cost of a pair of trousers was 55% the cost of a shirt. If Shubham intended to buy a pair of trousers and a shirt both, then how much money must he have, if the shirt costs ₹ 31.50 more than the pair of trousers?

A. ₹ 38.50

B. ₹ 70

C. ₹ 108.50

D. ₹ 120.55

---

43. A man sells two horses for ₹ 4000 each, neither losing nor gaining in the deal. If he sold one horse at a gain of 25%, then the other horse is sold at a loss of _____.

A. $33\frac{1}{3}\%$ B. $16\frac{2}{3}\%$

C. 50% D. 47%

---

44. A school committee invited 135 students to the annual function. If the ratio of number of boys to the girls was 4 : 5, then find the number of boys and girls invited by the committee.

A. 70, 65

B. 47, 88

C. 30, 105

D. 60, 75

---

45. Saumya borrowed ₹ 15000 at the rate of 12% p.a. for two years and another amount at the rate of 15% p.a. for two years. The total interest paid by her was ₹ 9000. Find the amount she borrowed at 15% p.a.

A. ₹ 31000

B. ₹ 33000

C. ₹ 28000

D. ₹ 18000

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46. Read the given statements carefully and select the correct option.

Statement-I : While dividing a whole number by a fraction, we always get an integer.

Statement-II : The product of a proper and an improper fraction is always less than both the improper fraction and the proper fraction.

A. Both Statement-I and Statement-II are true.
B. Both Statement-I and Statement-II are false.
C. Statement-I is true but Statement-II is false.
D. Statement-I is false but Statement-II is true.

47. Match the following:

	Expression	**Value of x**
(P)	$\frac{16^x \times 4^5 \times (64)^3}{(4)^2 \times (256)^4} = 256$	(i) 1
(Q)	$64^{2x+1} = (2)^{3x-3}$	(ii) 4
(R)	$\left(\frac{2}{3}\right)^{x-1} \times \left(\frac{81}{16}\right) = \frac{4}{9}$	(iii) 7
(S)	$5^x = (125)^{1/3}$	(iv) –1

A. (P) → (ii); (Q) → (i); (R) → (iv); (S) → (iii)
B. (P) → (i); (Q) → (ii); (R) → (iii); (S) → (iv)
C. (P) → (ii); (Q) → (iv); (R) → (iii); (S) → (i)
D. (P) → (iii); (Q) → (ii); (R) → (iv); (S) → (i)

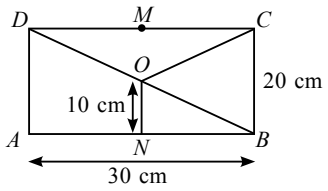
48. Read the given statements carefully and state 'T' for true and 'F' for false.

(i) Mean, Median and Mode can never be same for any data.
(ii) Median of the data can be less than the mean of data.
(iii) Mean of the observations is the sum of all the observations.

	(i)	**(ii)**	**(iii)**
A.	T	T	T
B.	T	F	T
C.	T	T	F
D.	F	T	F

49. $ABCD$ is a rectangle. If N, O and M are the mid points of AB, BD and CD respectively, then find :

(i) area of (ΔCOD)
(ii) area of ($CBNO$) : area of ($ABCD$)



	(i)	**(ii)**
A.	250 cm^2	2 : 1
B.	150 cm^2	1 : 4
C.	100 cm^2	3 : 2
D.	150 cm^2	3 : 8

50. Simplify :

(i) If the measure of two supplementary angles are $6x + 20°$ and $10x$, then find the value of x.
(ii) Find the complement angle of $\frac{3}{5} \times 45°$.
(iii) If two complementary angles are in the ratio 1 : 5, then find the smallest angle.

	(i)	**(ii)**	**(iii)**
A.	10°	63°	15°
B.	20°	27°	15°
C.	20°	63°	10°
D.	10°	27°	30°

SPACE FOR ROUGH WORK

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